



Optical Rain Gauge User Manual

(Type 485)

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1. Product Introduction

Product Description

This instrument is an optical rain sensor. It is a product that measures rainfall. It uses the principle of optical sensing to measure rainfall. There are multiple optical probes built in. The rain detection is reliable. It is different from the traditional mechanical rain sensor. The optical rain sensor has a smaller volume. More sensitive, reliable, smarter and easier to maintain.

It can be widely used in industries and fields such as smart irrigation, ship navigation, mobile weather stations, automatic doors and windows, and geological disasters.

Features

1. Small size, light weight and simple installation.
2. Low power consumption design, save energy
3. High reliability, can work normally in high temperature and high humidity environment
4. Easy maintenance design, not easy to be blocked by fallen leaves
5. Optical measurement, accurate measurement
6. RS485 signal output, long communication distance, convenient collection

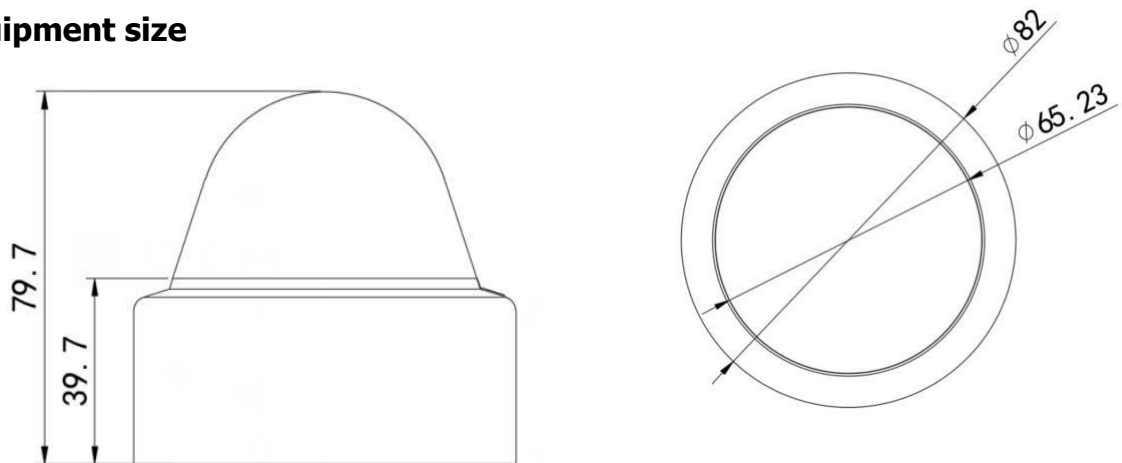
Main Specifications

Sense of rain diameter	6CM
Supply voltage	10~30V DC
Power consumption	Less than 0.24W (12V DC, current less 20mA)
Typical accuracy	±5%
Maximum instantaneous rainfall	24mm/min
output method	RS485 output
Operating temperature	-40~60°C
Working humidity	0~99%RH (no condensation)
Working pressure range	Standard atmospheric pressure ±10%
Resolution	Standard 0.1mm
Place of origin	China

Product Model

			Company code
GYL-			Optical rain gauge
	PL-		Pulse output
		1	-1 shell

Equipment size



2. Equipment installation instructions

Inspection before equipment installation

Equipment List

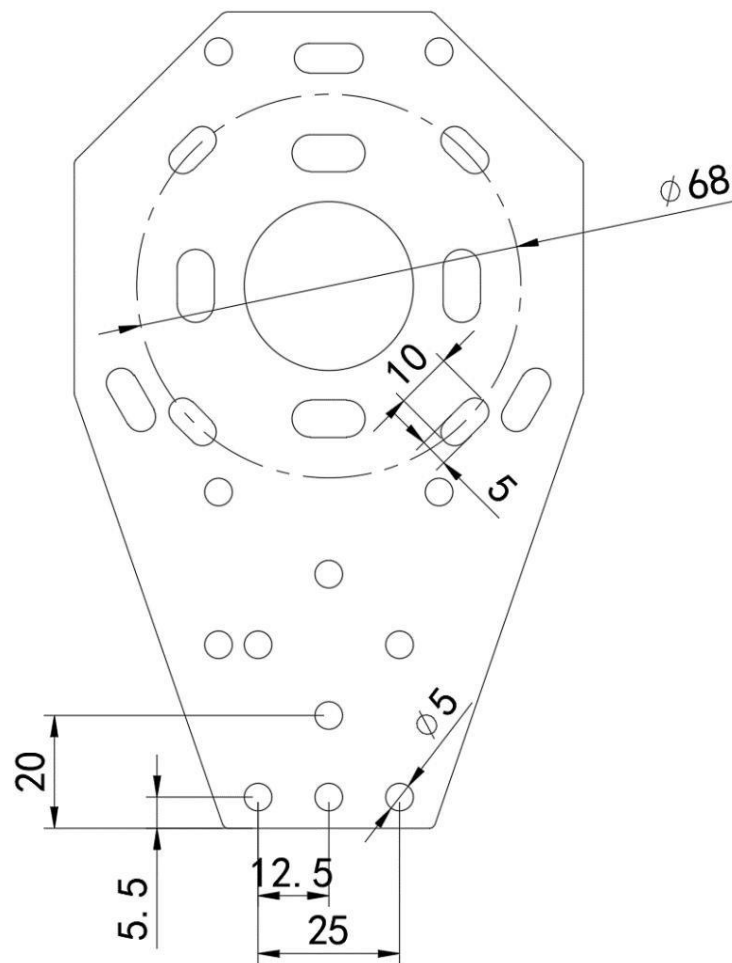
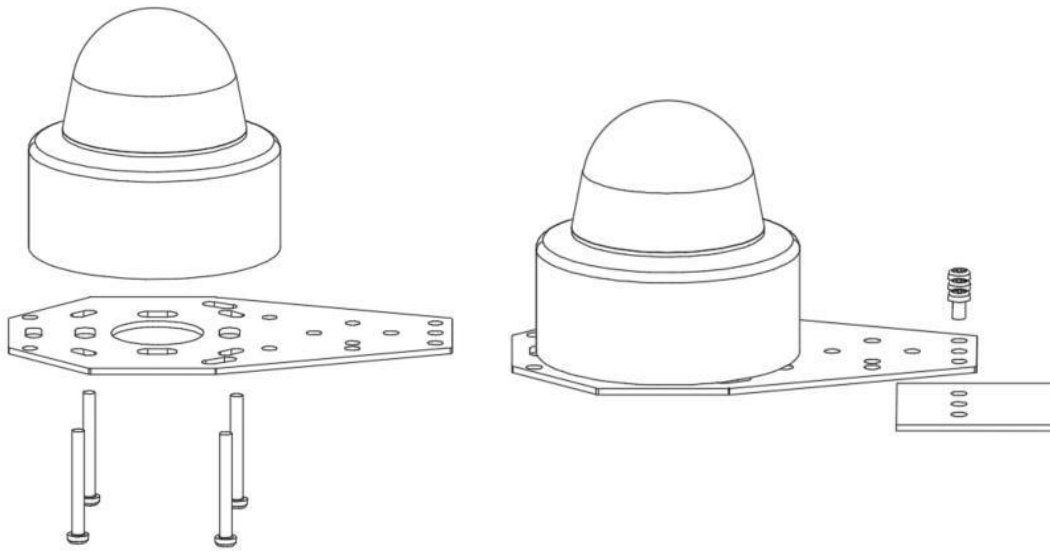
- Main equipment 1
- Tray 1
- 4 M4*35 screws
- M4*10 screws and nuts 3 each
- Warranty card, certificate of conformity 1 copy

Before installing the rain sensor, it must be ensured that the surface of the cover is dry. Any drop of water vapor will cause measurement errors. You can optionally use some desiccant in it.

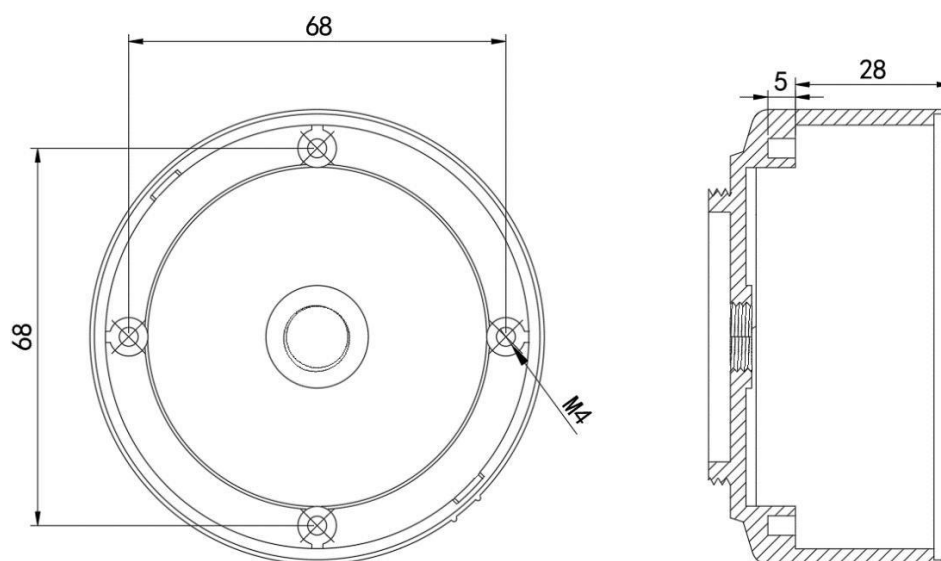
Check that the housing has been tightened, and the lower PG7 has been locked and not loose.

Outdoor installation and debugging

The rain sensor needs to be installed in an open place, and there should be no obstructions around and above it. Install the device on the bracket in the accessories first, and fix the device and the bracket with 4 M4*35 304 stainless steel screws and nuts. Then install the bracket to the position to be installed (the position to be installed requires a $\phi 5$ round hole). The bracket should be installed horizontally. Finally, the bracket and equipment are fixed by three M4*10 304 stainless steel screws and nuts.



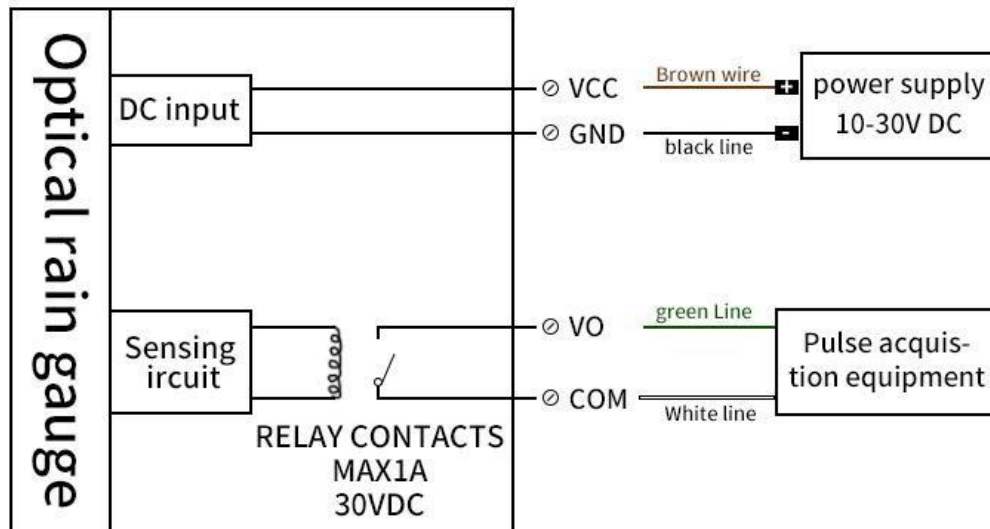
If the bracket is not used, the customer can install it according to the position and size of the copper stud in the figure below.



Wiring instructions

Thread color	Description
brown	Positive power supply (10~30V DC)
black	Power negative
Yellow	485 A
Blue	485 B

Wiring diagram



Note: This device is a dry contact pulse output, the maximum withstand voltage and current is 30VDC/1A

3.Configuration software installation and use

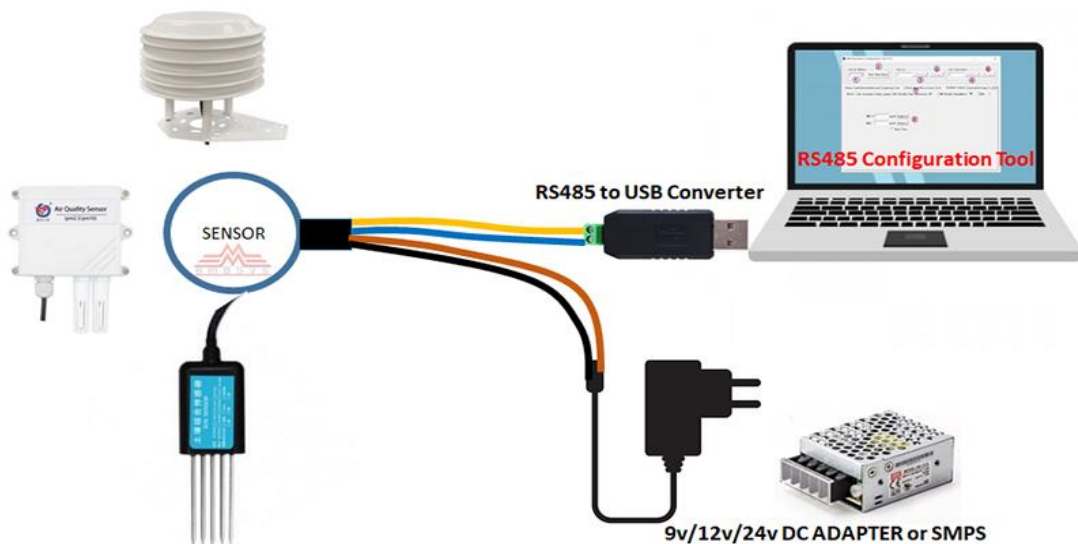
Tools Required

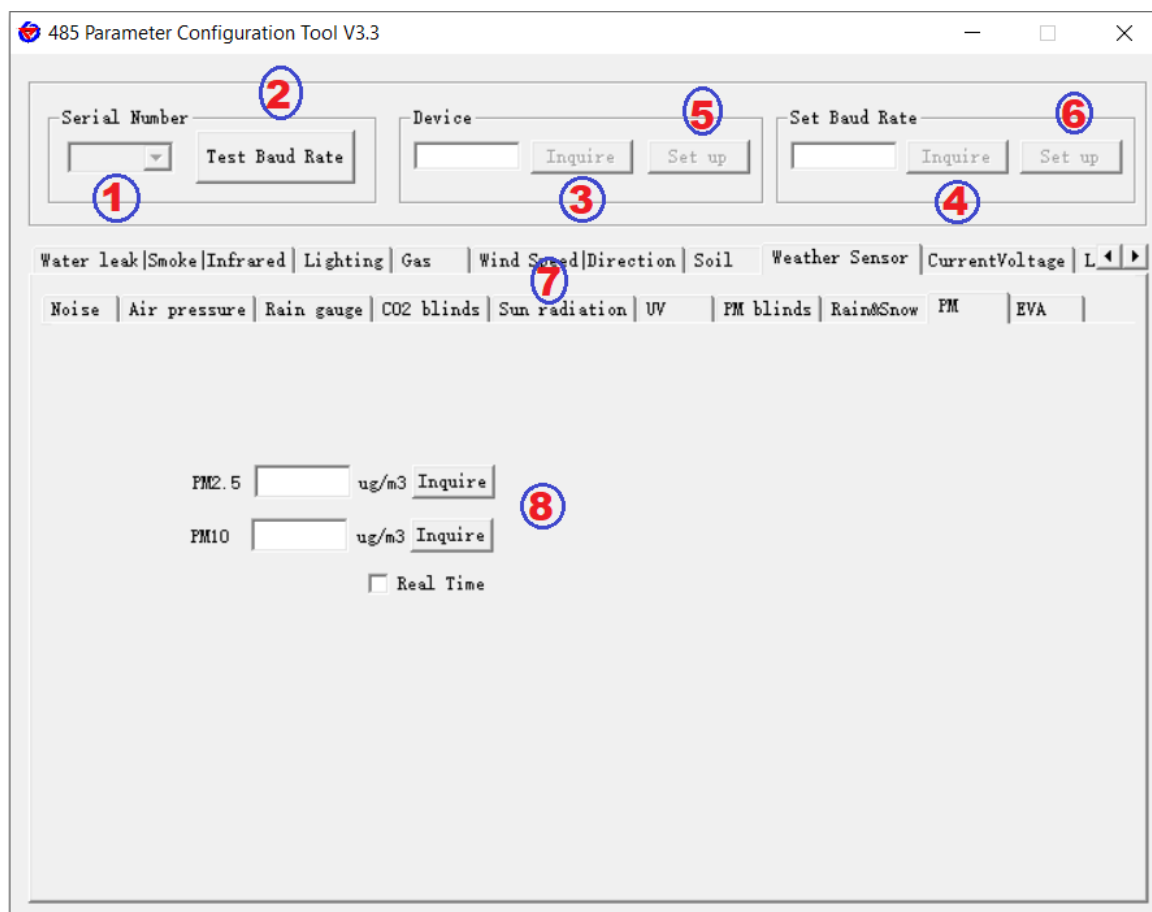
We need below tools to set or change the parameters of the sensors.

1. RS485 to USB Converter (To connect the sensor with your laptop. You should get a RS485 to USB converter and install corresponding driver. You can purchase from us too)
2. Power Supply Unit (To give power supply to the sensor)
3. RS485 Parameter Configuration Tool (You can download from our website or contact us)

Parameter settings

After Connected RS485-USB converter and Sensor and PC as shown in below image, open RS485 Parameter Configuration Tool. (Also you can check the sensor with moods can or similar software)





1. Choose CM port Serial Number
2. Click this button to test the baud rate
3. Click this button to check the Modbus device ID
4. Click this button to check the Baud rate of the device
5. Enter a value in the [Device] field and click this button to change the device ID
6. Enter a value in the [baud rate] field and click this button to change the baud rate
7. Choose your sensor from Main Tab and then Sub Tab
Click Inquire button to read the values from the sensor

4. Basic communication parameters

Code	8-bit binary
Data bit	8 digits
Parity bit	no
Stop bit	1 person
Error check	CRC (redundant cyclic code)
Baud rate	2400bit/s, 4800bit/s, 9600 bit/s can be set, the factory default is 4800bit/s

Data frame format definition

Adopt Modbus-RTU communication protocol, the format is as follows:

Initial structure ≥ 4 bytes of time

Address code = 1 byte

Function code = 1 byte

Data area = N bytes

Error check = 16-bit CRC code

End structure ≥ 4 bytes of time

Address code: is the address of the transmitter, which is unique in the communication network (factory default 0x01).

Function code: The instruction function of the command sent by the host. The transmitter can use function code 0x03 (read register data) 0x06, 0x10 (write register data). Data area: The data area is the specific communication data. Note that the 16-bit data high byte is in front!

CRC code: Two-byte check code.

Host inquiry frame structure:

address code	function code	Register start address	Register length	Check code low	Check code high
1 byte	1 byte	2 bytes	2 bytes	1 byte	1 byte

Slave response frame structure:

address code	function code	Effective number of bytes	Data area	Second data area	Nth data area	Check code
1 byte	1 byte	1 byte	2 bytes	2 bytes	2 bytes	2 bytes

Register address

Register address	content	Operation (hexadecimal)	Scope and definition
0000 H	Rainfall value	03/06	the rainfall value is 10 times larger than the actual value
07DOH	Device address	03/06	1~254 (factory default)
07D1H	Device baud rate	03/06	0 means 2400 1 means 4800 2 means 9600

Communication protocol example and explanation

Query the current rainfall value

Interrogation Frame:

address code	function code	starting address	Data length	Check code low	Check code high
0x01	0x03	0x00 0x00	0x00 0x01	0x84	0x0A

Response frame

address code	function code	Effective number of bytes	Baud rate	address	Check code low	Check code high
0x01	0x03	0x02	0x00 0x0A	0x00 0x01	0x38	0x43

Current rainfall value: (Uploaded value is expanded by 10 times)

000A (hexadecimal system) = 10 (decimal system) → rainfall value: 1.0mm
rainfall data through the following operations.

rainfall data through the following operations.

Clear rainfall data

address code	function code	starting address	Clear command	Check code low bit	Check code high
0x01	0x06	0x00 0x00	0x00 0x05A	0x09	0xF1

Response frame

Address code	function code	starting address	Clear command	Check code low bit	Check code high
0x01	0x06	0x00 0x00	0x00 0x05A	0x09	0xF1

Modify current address

Inquiry frame: (If the current address is 01, the address needs to be modified to 02)

address code	function code	starting address	Clear command	Check code low bit	Check code high
0x01	0x06	0x07 0xD0	0x00 0x02	0x08	0x86

Response frame

Address code	function code	starting address	Clear command	Check code low bit	Check code high
0x01	0x06	0x07 0xD0	0x00 0x02	0x08	0x86

Modify the current baud rate

Inquiry frame: (If the current baud rate is changed from 4800 to 9600)

address code	function code	starting address	Clear command	Check code low bit	Check code high
0x01	0x06	0x07 0xD1	0x00 0x02	0x59	0x46

Response frame

Address code	function code	starting address	Clear command	Check code low bit	Check code high
0x01	0x06	0x07 0xD1	0x00 0x02	0x59	0x46

Look for the address

When the user forgets the address, the following function codes can be used to query the address.

Interrogation frame:

address code	function code	starting address	Clear command	Check code low bit	Check code high
0xFF	0x03	0x07 0xD0	0x00 0x01	0x91	0x59

Response frame

Address code	function code	starting address	Clear command	Check code low bit	Check code high
0xFF	0x03	0x02	0x00 0x01	0x50	0x50

The address code read is the real address of the device:01

5. Common problems and solutions

Device cannot connect to PLC or computer

possible reason:

- 1) The computer has multiple COM ports, and the selected port is incorrect.
- 2) The device address is incorrect, or there is a device with a duplicate address (all the factory defaults to 1).
- 3) Baud rate, check mode, data bit, stop bit error.
- 4) The host polling interval and the waiting response time are too short and need to be set to more than 200ms.
- 5) The 485 bus is disconnected, or the A and B lines are reversed.
- 6) If the number of devices is too large or the wiring is too long, the power should be supplied nearby, add 485 enhancer, and increase the resistance of 120Ω terminal.
- 7) The USB to 485 driver is not installed or damaged.
- 8) Equipment damage.

6. Precautions

Daily maintenance

The instrument is outdoors for a long time and the use environment is very harsh. Therefore, the surface of the instrument should be kept clean and often wiped with a soft cloth. The instrument should generally be cleaned once a month for long-term operation and once every three months;

Use warning

Do not use the rain sensor in any place that will cause serious consequences due to the wrong sensor indication. It is the responsibility of the system designer or system integrator to ensure fault tolerance and not cause catastrophic consequences due to an error in one component (including the rain sensor). The company is not responsible for any related consequences caused by incorrect sensor indications.

7. Contact information

EmbSys Technologies Pvt Ltd
3/50, 2nd Cross Street
Sundar Nagar, Ekkattuthangal
Chennai, India. Pincode – 600032

For Purchase :	Visit: www.embsys.in e-Mail: sales@embsys.in Call/Whatsapp/Signal/Telegram +91 8148834200
For Technical Support :	e-Mail: support@embsys.in Call / Whatsapp +91 8148834211